

----- Active space -----

ABNORMALITIES OF HEAD SIZE AND SHAPE

Microcephaly

00:00:14

Indicative of small brain size (Head size).

Definitions :

Head circumference (HC) for age $< -2 \text{ SD} \rightarrow$ Low value.

True/severe/significant microcephaly : HC for age $< -3 \text{ SD} \rightarrow$ Pathological.

ETIOLOGY :

Primary :

Defective brain development.

1. Genetic defects :

- Trisomy (21, 18, 13).
- Cornelia de Lange syndrome.
 - Eyebrows are joined in the middle $\rightarrow$ Synophrys.
- Rubinstein taybi syndrome :
 - A/w broad thumb.

2. Familial microcephaly : Autosomal recessive inheritance.

Secondary :

External factors affecting brain growth.

1. Maternal factors :

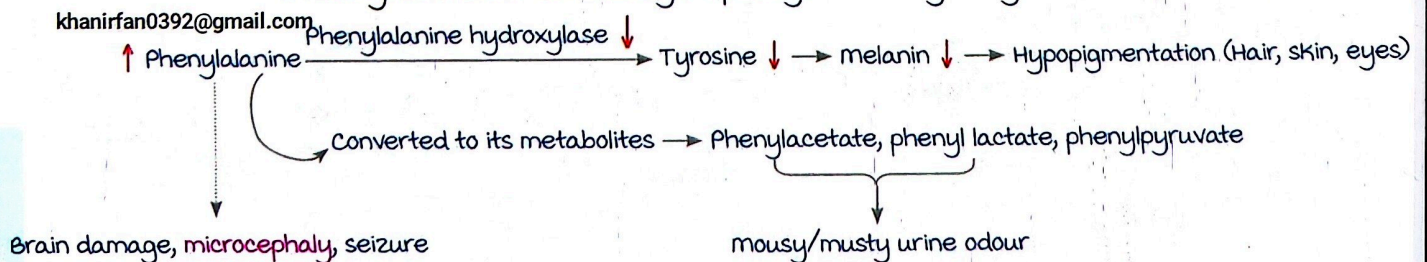
- TORCH infections.
- Toxins : Alcohol exposure.
- Drugs : Phenytoin.
- Radiation exposure during pregnancy.

2. Perinatal factors : Birth asphyxia.

3. Other CNS insults :

- Trauma/injury.
- Infections.

4. Phenylketonuria : Deficiency of phenylalanine hydroxylase.



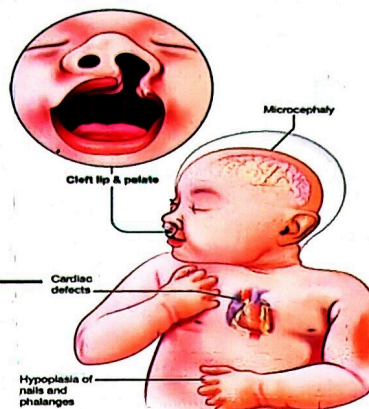
MICROCEPHALY D/T TERATOGENS :

1. Fetal alcohol syndrome :



microcephaly

2. Phenytoin (Diphenyl hydantoin) exposure :



most commonly : ventricular septal defect

Fetal hydantoin syndrome

LATE ONSET/ACQUIRED SYNDROMES :

Noticed during development of the child (Not at birth).

1. Rett syndrome :

- X-linked dominant inheritance.
- MECP-2 gene defect.
- Normal head size → microcephaly after 1 yr.
- Features :
 - Developmental regression : Loss of previously acquired milestones.
 - Hand wringing stereotypic movements : Characteristic (Abnormal hand movements in the midline).
 - Speech defects.
 - Progressive ataxia → wheelchair dependent by adolescence.

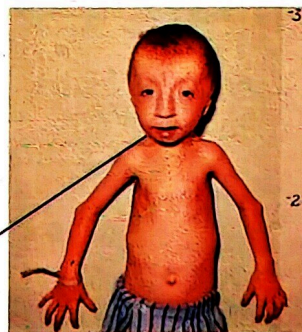


Abnormal midline hand movements

2. Seckel syndrome :

Bird like face + beak like nose

3. Angelman syndrome



Seckel syndrome

Macrocephaly

00:16:23

Definition :

HC for age $> +2$ SD.**ETIOLOGY :**

fluid in the brain :

Hydrocephalus.

Hydranencephaly : Complete replacement of brain tissue with fluids.

- Transillumination positive.

----- Active space -----

2. Thick skull :

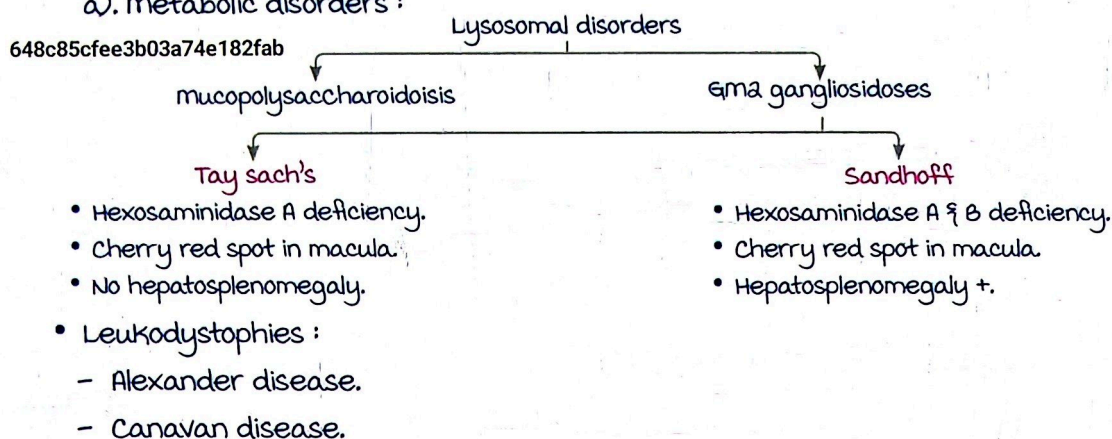
- Hemolytic anemia → Extramedullary hematopoiesis → Expansion of the skull.
Eg : β -thalassemia major.

- Achondroplasia.

- Osteopetrosis.

3. ↑ Brain volume : megalencephaly.

a). metabolic disorders :



b) Genetic syndromes :

- Sotos syndrome/cerebral gigantism.
- NF type I.
- **Fragile X syndrome** : D/t CGG trinucleotide repeats.
 - Long face, large ears.
 - **macroorchidism** (Enlarged testis).

c) Non syndromic :

Familial megalencephaly → m/c type & cause of megalencephaly.

Craniosynostosis

00:25:28

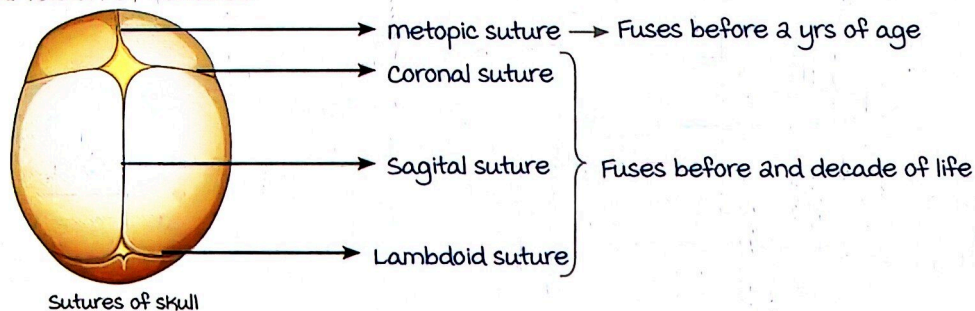
Abnormalities of head shape :

Cranio : Skull.

Synostosis : Fusion of sutures.

} Premature fusion of the sutures

Normal fusion of sutures :



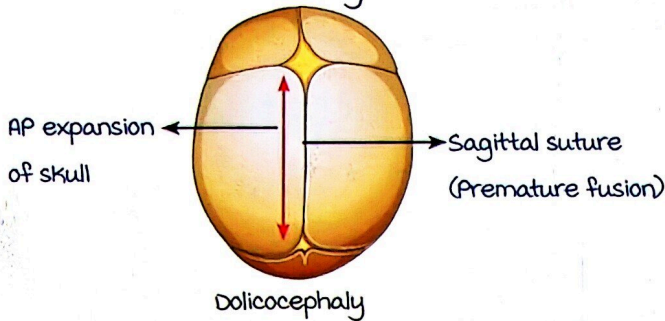
Rule : Suture fusion → Skull growth arrests perpendicular to suture line.

- Skull growth occurs parallel to fused suture.

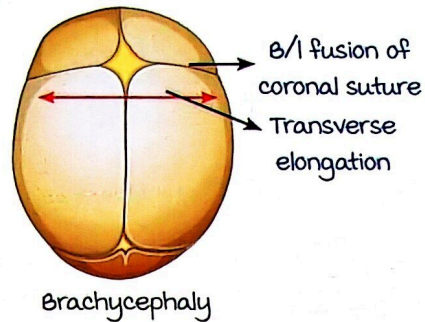
TYPES

1. Dolicocephaly/scaphocephaly:

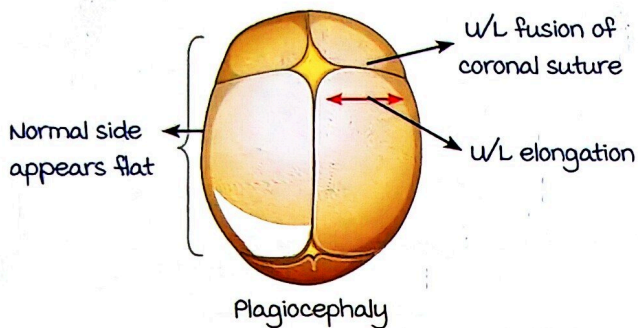
m/c suture involved in craniosynostosis.



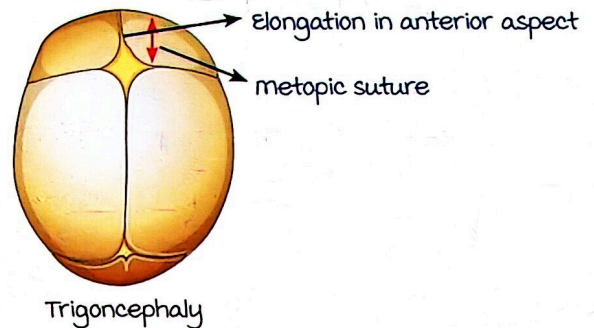
2. Brachycephaly:



3. Plagiocephaly:



4. Trigonocephaly:



5. Oxycephaly/Turricephaly:

Premature fusion of multiple sutures → Irregular shape of skull.

ASSOCIATED GENETIC SYNDROMES

	Crouzon syndrome	Apert syndrome
Inheritance	Autosomal dominant	Autosomal dominant
Craniosynostosis	Brachycephaly	Turricephaly (Tower shaped skull)
Features	- Exophthalmos - Hypertelorism - maxillary hypoplasia D/t shallow orbit	- Exophthalmos (Not prominent) - Syndactyly (Fused fingers), mitten hand deformity
IQ	Normal IQ	Low IQ



Crouzon syndrome

Apert syndrome

Carpenter syndrome: Autosomal recessive.

Similar to apert syndrome.

A/w obesity & hypogonadism.

